DASC 45th Digital Avionics Systems Conference
Orlando, Florida, USA · September 13–17, 2026

Stochastic Strategic Demand–Capacity Balancing Using Model-Free Reinforcement Learning

Lucas Orbolato Carvalho · Mayara Condé Rocha Murça
Aeronautics Institute of Technology (ITA) — São José dos Campos, Brazil
orbolatoloc@fab.mil.br
mayara@ita.br

KEY RESULT RL reduces tactical congestion cost with minimal intervention: 13.3% at SBGR and 64.7% in the UAM case study.

01 MOTIVATION

Weather is the dominant driver of demand-capacity imbalance

Weather uncertainty makes strategic demand–capacity balancing difficult. **Deterministic ground-holding schemes** cannot fully adapt to stochastic capacity variations, while UAM adds short flights, limited vertiport parking, and coupled arrival–departure constraints.

74.3%
of U.S. system-impacting delays over 15 min are weather-related
FAA, 2017–2023

20–50%
of European ATFM delay is attributed to weather
EUROCONTROL, 2024

40.6%
of ATFM measures in Brazilian airspace due to weather
DECEA, 2025

02 OBJECTIVES

01 Integrated RL framework

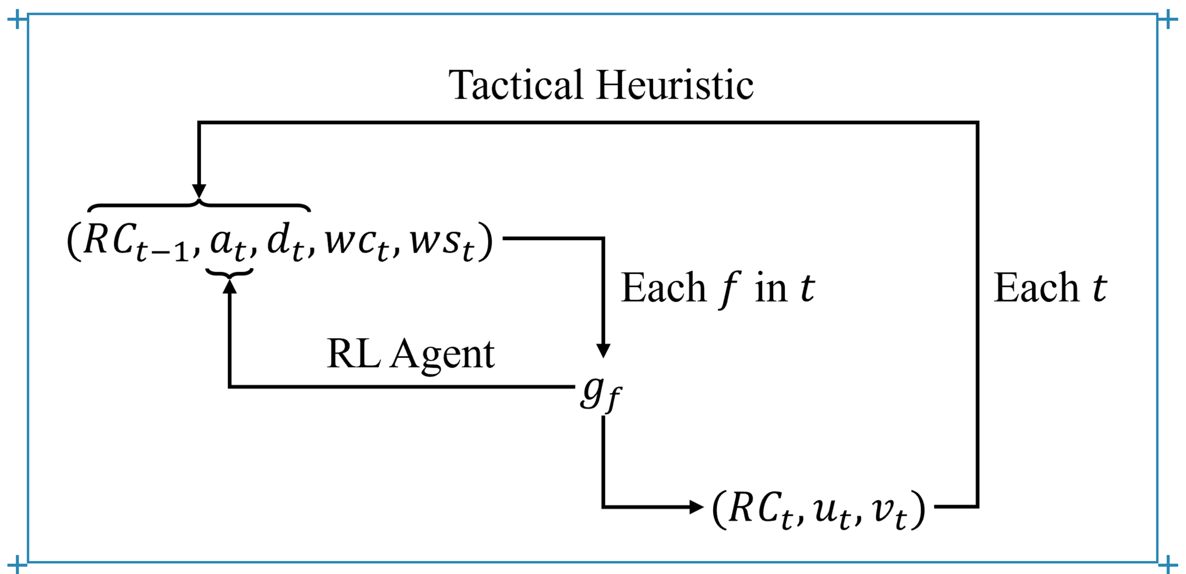
Develop a reinforcement learning framework for demand-capacity balancing that integrates strategic ground-delay allocation with tactical capacity management under uncertain weather forecasts.

02 Two operational paradigms

Demonstrate the applicability of the framework, without modification, to both conventional ATFM at São Paulo/Guarulhos and UAM traffic management in the São Paulo Metropolitan Region.

03 METHOD

One loop, two decision layers — an MDP on 15-minute epochs



Strategic layer — RL agent
For each arrival flight *f* in epoch *t*, choose a ground delay g_f in 15-minute increments, up to one hour.

Tactical layer — heuristic
Given the wind state, select a feasible runway configuration (or vertiport side) and set arrival/departure service rates from the capacity envelope.

Coupling
Reallocated demand changes future queues, so the tactical consequence of each delay is fed back into the reward.

STATE

$$s_t = (\,RC_{t-1}, a_t, d_t, wc_t, ws_t\,)$$

Previous configuration · arrival queue · departure queue · weather category (VMC / IMC / LIMC) · wind state, encoding which sides stay usable at CW ≤ 20 kt and TW ≤ 5 kt. Horizon T = 96 epochs.

ACTION, MASKED

$$g_f \in \{0, 1, 2, 3, 4\} \times 15 \text{ min}$$

Capped at one hour and at the time remaining in the horizon; a binary mask removes infeasible delays so the agent never explores them.

TACTICAL SERVICE RATES

$$u_t = \min(\,u_{max}[wc_t, RC_t], a_t\,)$$
$$v_t = \min(\,v_{max}[wc_t, RC_t, u_t], d_t\,)$$

Configuration taken from the operationally valid set $RC_t \in \mathcal{RC}(ws_t)$.

REWARD — THE TACTICAL CONSEQUENCE

$$J^{tac} = \sum_{t=0}^{T} [\,\alpha(a_t - u_t)^2 + (d_t - v_t)^2 + \delta \cdot 1(RC_t \neq RC_{t-1}) \cdot (\alpha a_t^2 + d_t^2)\,]$$
$$R_t = -(\,\Delta J^{tac}(t) + \eta \sum_f g_f\,), \quad \eta = 0.1$$

A delay may relieve congestion in one epoch while worsening another, so the reward carries the full change in tactical cost over the remaining episode. Evaluation metric: €45 per delayed departure minute and €90 per airborne arrival minute, following EUROCONTROL (2024).

1. WEATHER DYNAMICS

$$P(\,METAR_t = j \mid METAR_{t-1} = i\,)$$

Empirical transition frequencies between VMC, IMC and LIMC at 15-min resolution.

2. FORECAST ERROR

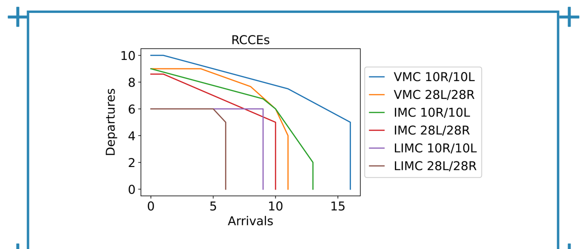
$$P(\,METAR = j \mid TAF = i\,)$$

Confusion matrices estimated by comparing forecasts against observed realizations.

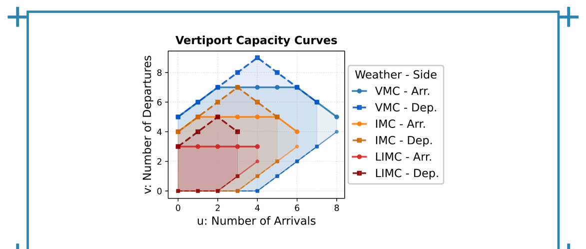
TRAINING-TIME WEATHER SAMPLING

$$wc_t \sim P(wc_t \mid wc_{t-1}, \hat{wc}_t) \propto P(wc_t \mid wc_{t-1}) \cdot P(wc_t \mid \hat{wc}_t)$$

The wind state is sampled the same way, after converting wind speed and direction into crosswind and tailwind components relative to runway orientation.



Runway envelopes at SBGR, fitted by quantile regression (Ramanujam & Balakrishnan, 2009): concave and invertible.



Vertiport curves. Limited parking imposes a **minimum** departure count per arrival level, so the envelope is non-invertible.

04 CASE STUDIES

Train on forecast, test on observation — 30 test days per case

ALGORITHM
Maskable PPO — Stable Baselines 3 · Gymnasium; 500,000 training episodes per day (conventional), 300,000 (UAM)

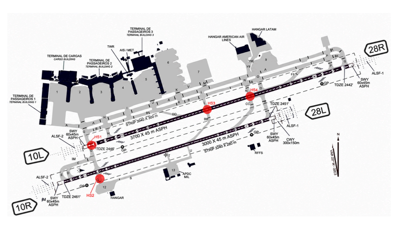
DATA
24 months of METAR/TAF (REDEMET) at 15-min resolution; ANAC SIROS schedules corrected with ANAC VRA and FlightRadar24 tracks. TAF for training, METAR for testing.

BASELINE — NO RESCHEDULING
The same tactical heuristic and capacity model with $g_f = 0$ for every flight. The strategic rescheduling cost is omitted in evaluation, so the metric isolates tactical congestion reduction.

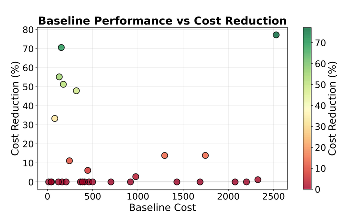
GROUND DELAY ALLOCATED — 10,144 / 2,857 FLIGHTS

DELAY	SBGR	UAM
0 slots	**98.9%**	**95.6%**
1 slot (15 min)	0.7%	2.2%
2 slots (30 min)	0.4%	1.8%
3 slots (45 min)	0.0%	0.2%
4 slots (60 min)	0.0%	0.3%

CASE 1 — CONVENTIONAL ATM AT SBGR



300k+ annual operations on two parallel runways; configurations 10R/10L and 28L/28R.



Daily percentage reduction against baseline cost, 30 test days.

13.3%
aggregate mean-cost reduction

12/30
days improved

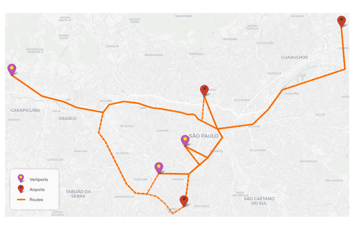
77.2%
best single day

0
days degraded

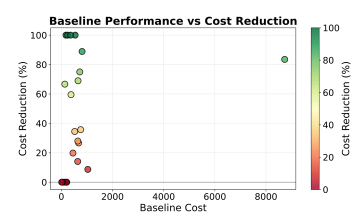
1.1%
of flights delayed

The policy matches the baseline on most days and delivers large gains on a minority, including ~77% on the most congested day. Wilcoxon signed-rank **p = 0.002183**. A conservative policy: 98.9% of flights fly as scheduled, and delay is spent only where it removes real tactical congestion.

CASE 2 — UAM TRAFFIC MANAGEMENT, RMSP



Three vertiport candidates in the RMSP, routed to SBGR, SBSP and SBMT over the existing REH corridors.



Daily percentage reduction against baseline cost, 30 test days.

64.7%
aggregate mean-cost reduction

23/30
days improved

43.9%
mean daily reduction

83.5%
worst-case cost cut

4.4%
of flights delayed

Widespread reductions, several complete eliminations of the daily tactical cost, and ~83% on the most congested day; maximum daily cost falls from €8,730 to €1,440. Wilcoxon signed-rank **p = 0.000281**. The synthetic UAM baseline is deliberately not pre-optimized against vertiport capacity, leaving more headroom than an operationally planned schedule.

05 FUTURE PERSPECTIVES

What the framework contributes, and what it does not yet cover

CONTRIBUTIONS

01 Model-free RL framework for strategic DCB with probabilistic weather representation
02 Integrated strategic and tactical optimization of delay, configuration and service rates
03 Action masking that removes infeasible ground-delay exploration and accelerates learning
04 Demonstrated transfer across conventional ATM and UAM with the same algorithm

BEYOND A SINGLE AIRPORT
Network effects, airline recovery, passenger connections and multi-airport coordination remain unexplored, as do wind-state definitions at other aerodromes.

WEATHER REPRESENTATION
Route-level low-altitude weather and finer variables would improve realism over VMC/IMC/LIMC categories derived from airport METAR and TAF.

Selected references: Richetta & Odoni (1994); Ramanujam & Balakrishnan (2009); Bertsimas et al. (2011); Estes & Ball (2020); Shone et al. (2021); Jones et al. (2023); Carvalho & Murça (2025); Sarkar et al. (2025); EUROCONTROL (2024).
Acknowledgements: Brazilian Civil Aviation Secretariat (SAC), Decentralized Execution Agreement (TED) No. 11525720240005-003882/2024, and the Brazilian Air Force (FAB).

Paper page
Paper 276
orbolato.github.io/dasc2026-paper-276

45TH DIGITAL AVIONICS SYSTEMS CONFERENCE
ORLANDO, FL · SEPT 2026